DESCENT OF INVERTIBLE MODULES ALONG LIMITS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Author omission kept verbatim: 'After increasing i we may assume this map is surjective (details omitted; hint: use Algebra, Lemma \ref{algebra-lemma-module-map-property-in-colimit} on affine open cover)'. The omission is auxiliary (a lift existence step) and carries a hint plus a named proved lemma.

Allowed standard prerequisites (editorial).

- Module map property in colimit. Algebra, Lemma algebra-lemma-module-map-property-in-colimit

Source reference. Original lemma 01ZR: sources/stacks-chow/tags.txt, source lines 2604–2604. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02JO: sources/stacks-chow/tags.txt, source lines 3343–3343. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NX: sources/stacks-chow/tags.txt, source lines 875–875. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B8K: sources/stacks-chow/tags.txt, source lines 14143–14143. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05LI: sources/stacks-chow/tags.txt, source lines 7237–7237. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $S = \lim S_i$ be the limit of a directed system of quasi-compact and quasi-separated schemes S_i with affine transition morphisms. Then*

1. *any finite locally free $\mathcal{O}_S$-module is the pullback of a finite locally free $\mathcal{O}_{S_i}$-module for some i,*
2. *any invertible $\mathcal{O}_S$-module is the pullback of an invertible $\mathcal{O}_{S_i}$-module for some i, and*
3. *any finite type quasi-coherent ideal $\mathcal{I} \subset \mathcal{O}_S$ is of the form $\mathcal{I}_i \cdot \mathcal{O}_S$ for some i and some finite type quasi-coherent ideal $\mathcal{I}_i \subset \mathcal{O}_{S_i}$.*

Proof. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_S$-module. Since finite locally free modules are of finite presentation we can find an i and an $\mathcal{O}_{S_i}$-module $\mathcal{E}_i$ of finite presentation such that $f_i^*\mathcal{E}_i \cong \mathcal{E}$, see Lemma 01ZR. After increasing i we may assume $\mathcal{E}_i$ is a flat $\mathcal{O}_{S_i}$-module, see Algebra, Lemma 02JO. (Using this lemma is not necessary, but it is convenient.) Then $\mathcal{E}_i$ is finite locally free by Algebra, Lemma 00NX.

If $\mathcal{L}$ is an invertible $\mathcal{O}_S$-module, then by the above we can find an i and finite locally free $\mathcal{O}_{S_i}$-modules $\mathcal{L}_i$ and $\mathcal{N}_i$ pulling back to $\mathcal{L}$ and $\mathcal{L}^{\otimes -1}$. After possible increasing

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i we see that the map $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes -1} \to \mathcal{O}_X$ descends to a map $\mathcal{L}_i \otimes_{\mathcal{O}_{S_i}} \mathcal{N}_i \to \mathcal{O}_{S_i}$. And after increasing i further, we may assume it is an isomorphism. It follows that $\mathcal{L}_i$ is an invertible module (Modules, Lemma 0B8K) and the proof of (2) is complete.

Given $\mathcal{I}$ as in (3) we see that $\mathcal{O}_S \to \mathcal{O}_S/\mathcal{I}$ is a map of finitely presented $\mathcal{O}_S$-modules. Hence by Lemma 01ZR this is the pullback of some map $\mathcal{O}_{S_i} \to \mathcal{F}_i$ of finitely presented $\mathcal{O}_{S_i}$-modules. After increasing i we may assume this map is surjective (details omitted; hint: use Algebra, Lemma 05LI on affine open cover). Then the kernel of $\mathcal{O}_{S_i} \to \mathcal{F}_i$ is a finite type quasi-coherent ideal in $\mathcal{O}_{S_i}$ whose pullback gives $\mathcal{I}$. $\square$